

Theory of Computation

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Decision Problems

- Generally, a computer program computes a function, say f , by
 - reading an input x from the domain of f
 - producing the output $f(x)$
- Let us first consider a very simple kind of functions, called *decision problems*, where the domain is N and the range is $\{0,1\}$, i.e. $f(x)$ is either 0 or 1. For example,

$$f_{\text{odd}}(x) = \begin{cases} 1 & \text{if } x \text{ is odd} \\ 0 & \text{otherwise} \end{cases}$$

- This kind of functions can also be represented by a set:
$$S = \{x \in \text{dom}(f) \mid f(x) = 1\}$$
- The above function f_{odd} can be represented the set of all odd natural numbers.

Decision Problems

- Yet another way to represent a decision problem f is by an infinite sequence of 0 and 1:

$$f(0), f(1), f(2), \dots$$

- For example, the function f_{odd} can be represented by

$$f_{odd}(0), f_{odd}(1), f_{odd}(2), \dots$$

$$0, 1, 0, 1, 0, 1, 0, 1, \dots$$

Decision Problems

- Given such a decision problem f , the question is

“Is there a computer program that computes it?”
- Suppose there is a computer program that can compute f , we say that f is computable or decidable.
- Alternatively, suppose S is the set representing f . We say that S is a computable or decidable set, if and only if there is a computer program that can determine for each $x \in \mathbb{N}$, whether x is in S or not.
- Are the following sets computable?
 - The set of all odd natural numbers
 - The set of all natural numbers divisible by 3
 - The set of all prime numbers

Decision Problems

- Is every decision problem computable?
- Unfortunately, no. We can show that there are more decision problems than there are computer programs.
- First, we recall how we compare the sizes of infinite sets.
- **Definition.** Two sets A and B are said to be **equinumerous** (i.e. having the same size), written $|A| = |B|$, iff there exists a bijection between the two sets.
 - For example, suppose Odd is the set of all odd natural numbers. We can show that $|N| = |Odd|$.
- **Definition.** A set is said to be **finite** iff the number of elements equals some natural number. A set that is not finite is said to be infinite.
- **Definition.** A set is said to be **countable** iff it is either finite or is equinumerous to the natural numbers.

Decision Problems

- Assumptions.
 1. A computer program is a binary representation of some natural number.
 2. Two different computer programs may compute the same decision problem. But two different decision programs cannot be computed by the same program.
- It follows from the first assumption that the set of all computer programs is countable.

Decision Problems

- Theorem. There exists some decision problem that is not computable.
- **Proof.** By contradiction. Assume that every decision problem is computable. Let $P_0, P_1, P_2, \dots$ be all the computer programs and let $f_0, f_1, f_2, \dots$ be the decision functions computed by each of those programs, respectively. Hence, the list $f_0, f_1, f_2, \dots$ includes all computable functions.
- Let us define the function f^* as follows:
$$f^*(n) = \begin{cases} 0 & \text{if } f_n(n) = 1 \\ 1 & \text{if } f_n(n) = 0 \end{cases}$$
- By this definition, f^* differs from each f_n when the input is n , i.e. $f^*(n) \neq f_n(n)$.
- But since every computable decision function is included in $f_0, f_1, f_2, \dots$, it follows that f^* is *not* computable.